RELATIVE LERAY SPECTRAL SEQUENCE IN QCOH FOR QCQS MORPHISMS

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Relative Leray spectral sequence. Cohomology on Sites, Lemma sites-cohomology-lemma-relative-Leray. Module-level spectral sequence.
- Q-adjunction defect is parasitic. Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-adjoint-kernel-parasitic (2). Kernel/cokernel parasitic.
- Higher pushforwards preserve parasitic modules. Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-pushforward-parasitic (2). R^p g_* keeps the defect parasitic.

Source reference. Original reference 0734: sources/stacks-chow/tags.txt, source lines 9048–9048. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original proposition 0771: sources/stacks-chow/tags.txt, source lines 9186–9186. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0779: sources/stacks-chow/tags.txt, source lines 9194–9194. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0775: sources/stacks-chow/tags.txt, source lines 9190–9190. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ and $g : \mathcal{Y} \to \mathcal{Z}$ be quasi-compact and quasi-separated morphisms of algebraic stacks. Let $\mathcal{F}$ be a quasi-coherent sheaf on $\mathcal{X}$. Then there exists a spectral sequence with E_2-page*

$$E_2^{p,q} = R^p g_{QCoh,*}(R^q f_{QCoh,*}\mathcal{F})$$

converging to $R^{p+q}(g \circ f)_{QCoh,*}\mathcal{F}$.

Proof. By Cohomology on Sites, Lemma 0734 the Leray spectral sequence with

$$E_2^{p,q} = R^p g_*(R^q f_*\mathcal{F})$$

converges to $R^{p+q}(g \circ f)_*\mathcal{F}$. By the results of Proposition 0771 all the terms of this spectral sequence are objects of $LQCoh^{fbc}(\mathcal{O}_\mathcal{Z})$. Applying the exact functor

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$Q_\mathcal{Z} : LQCoh^{fbc}(\mathcal{O}_\mathcal{Z}) \to QCoh(\mathcal{O}_\mathcal{Z})$ we obtain a spectral sequence in $QCoh(\mathcal{O}_\mathcal{Z})$ covering to $R^{p+q}(g \circ f)_{QCoh,*}\mathcal{F}$. Hence the result follows if we can show that

$$Q_\mathcal{Z}(R^p g_*(R^q f_*\mathcal{F})) = Q_\mathcal{Z}(R^p g_*(Q_\mathcal{X}(R^q f_*\mathcal{F}))$$

This follows from the fact that the kernel and cokernel of the map

$$Q_\mathcal{X}(R^q f_*\mathcal{F}) \longrightarrow R^q f_*\mathcal{F}$$

are parasitic (Lemma 0779) and that $R^p g_*$ transforms parasitic modules into parasitic modules (Lemma 0775). $\square$